

Name: Mejia, Dougherty, Melson, Kidwell

Date: 02-09-24

Class: 8

MyDesign Portfolio

Element C Initial Template

How to use this template

Element C: Presentation and Justification of Solution Design Requirements

Design Requirement #1 (**Safe**):

- Safety is the biggest concern. Student safety is always a number one priority when dealing with classroom management.
- We have to ensure that our design is safe for students to use and will not pose a threat to the safety of the students.
- A poorly designed filter falling on another student would lead to a lawsuit and reflect badly on the school as a whole
- In our prior research we determined that safety is of utmost importance and is ultimately the determining factor of whether or not a design can be implemented.

-Comments: Yes, safety is of the highest concern.

Design Requirement #2 (**Functional**)

- Our air filter must successfully carry out the overall intended function of filtering toxic vapors in Mrs. Poole's classroom.
- The customer intends for our air filter to effectively diminish the amount of toxic fumes and vapors from the classroom.
- Therefore, in order for a significant reduction of such vapors to occur, our filter *must* be designed in a manner that can remove and contain them from the air (i.e. use of fans, layered filters, etc.).

-Comments: The machine operating at peak efficiency will be very important.

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Products to consider when choosing a filter type: resin, PLA, ABS, wood burning. Will the filter successfully remove the VOCs or other toxins/smells in the air.

Design Requirement #3 (Space Effective)

- Our design must be small enough to fit in the space we were given, and be mounted to the wall or hung from the ceiling.
- When interviewing Mrs. Poole, we found out that her room is 15x25x10 and does not have any floor space for a filter.
- If our filter is too big or not mounted correctly, it is unusable in Mrs. Poole's room.

-Comments: Need a mounting system that is secure but still allows for maintenance

Floor space is at a premium so mounting to the wall or ceiling would be the best solution.

Design Requirement #4 (Easy Maintenance)

- Easy management was something asked of every stakeholder, it's a requirement because of the extensive responsibilities of the stakeholders leaving not much time to upkeep the filter.
- Having a complicated process to upkeep the device will discourage further implementation in other rooms
- When we interviewed Mr Buffon, we determined that easy maintenance is a crucial element of the filter's continued use.

-Comments: Also, a machine students can safely operate and maintenance is important

Design Requirement #5 (Durable/Longevity)

- The filter needs to have an appropriately long life span to prevent having to make/buy several new ones in a short amount of time.
- Just like any filter, our filter should last for a while, at least 2 years

-Comments: Ideally the filter should not have to be changed out more than 4 times a semester.

Design Requirement #6 (Easy to operate)

- This filter needs to be easy enough to operate so that students can turn it on and use it easily.

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- **Most of the filters we researched were operated with a simple switch or on button. We hope to replicate this level of ease in order to produce the most convenient air filter to our clients.**

-Comments: no comments on this particular design requirement

Design Requirement #7 (Quiet)

- **This filter needs to be quiet enough so that the teacher can teach clearly to the students without too much noise.**
- **Public feedback on prior experiments based on air filters shows that disturbing volumes of noise coming from the filter are one of the most apparent complaints.**
- **When we interviewed Mrs. Poole, one of her concerns was that our filter was quiet so as not to disturb her students in their work.**
- **Therefore, our filter must be as quiet as possible in order to maintain an effective classroom environment, both in air quality and instruction.**

-Comments: I do not want to feel like I am competing with the filter when I am talking to the class.

Also to consider:

8. Electric Powered

9. Aesthetic

